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Unit – 5 Inner Product Space (Lecture Note)

Introduction

→ Inner Product Spaces extend the concept of vectors and vector operations beyond ordinary Euclidean geometry.

→ In elementary mathematics, vectors are studied using the dot product, which allows us to measure lengths, distances, and angles. Inner Product Spaces generalize these ideas to more abstract vector spaces such as polynomial spaces, matrix spaces, and spaces of continuous functions.

→ The concept of an inner product provides a powerful tool for defining important geometric concepts such as norm, distance, orthogonality, and projection.

→ In this unit, we study inner products and their properties, norms and distances induced by inner products, orthogonal and orthonormal sets, orthogonal projections, and the Gram–Schmidt orthogonalization process. We also explore orthogonal matrices, QR decomposition, and least squares solutions.

→ These concepts play a fundamental role in mathematics, engineering, computer science, data analysis, machine learning, signal processing, computer graphics, and numerical computations.

Method 1 ⇝ Real Inner Product Space

Real Inner Product Space

→ Let V be a real vector space. A function $\langle \cdot , \cdot \rangle : V \times V \to \mathbb{R}$ is said to be inner product on V if the following conditions are satisfied.

For $u, v, w \in V$ and $k \in \mathbb{R}$

1	$\langle u, v \rangle = \langle v, u \rangle$	(Symmetry)
2	$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$	(Additivity)
3	$\langle ku, v \rangle = k\langle u, v \rangle$	(Homogeneity)
4	$\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0$ if and only if $u = 0$	(Positive Definiteness)

→ A real vector space V with an inner product i.e., $(V, \langle \cdot , \cdot \rangle)$ is called a real inner product space.

→ **Remark:**

We simply write inner product space instead of real inner product space.

Euclidean Inner Product

→ Let $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ be vectors of $\mathbb{R}^n$.

The function $\langle u, v \rangle = u \cdot v = u_1v_1 + u_2v_2 + \cdots + u_nv_n$ is called the Euclidean inner product, since it satisfies all the conditions of an inner product.

The Euclidean inner product is also known as the standard inner product OR dot product.

Weighted Euclidean Inner Product

→ Let $u = (u_1, u_2, \dots, u_n)$ and $v = (v_1, v_2, \dots, v_n)$ be vectors of $\mathbb{R}^n$, and let $w_1, w_2, \dots, w_n$ be positive real numbers called weights.

The function $\langle u, v \rangle = w_1u_1v_1 + w_2u_2v_2 + \cdots + w_nu_nv_n$ is called the weighted Euclidean inner product in $\mathbb{R}^n$ with weights $w_1, w_2, \dots, w_n$, since it satisfies all the conditions of an inner product.

Examples of Method-1: Real Inner Product Space

C	1	Let $u = (u_1, u_2)$ and $v = (v_1, v_2)$ be vectors in $\mathbb{R}^2$. Prove that weighted Euclidean inner product $\langle u, v\rangle = 3u_1v_1 + 2u_2v_2$ satisfies the four inner product axioms.
C	2	Let $V = \mathbb{R}^2$ and $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}, v = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ be vectors in V. If $\langle u, v\rangle = 3u_1v_1 + u_1v_2 + 4u_2v_1 + 5u_2v_2$ then check that $(\mathbb{R}^2, \langle , \rangle)$ is inner product space or not. **Answer: No**
C	3	Let $V = M_n(\mathbb{R})$ be the vector space of all $n \times n$ matrices with real entries. Prove first three conditions of inner product for the function defined by $\langle A, B\rangle = \text{tr}(A^T SB)$. Where S is fix symmetric matrix of order $n \times n$.

Method 2 ⇝ Properties of Inner Product and Norm

Properties of inner product

→ Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Consider $u, v, w \in V$ and $k \in \mathbb{R}$, then

1 $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$

2 $\langle u, kv \rangle = k\langle u, v \rangle$

3 $\langle u, 0 \rangle = \langle 0, u \rangle = 0$

4 $\langle -u, -v \rangle = -\langle -u, v \rangle = -\langle u, -v \rangle = \langle u, v \rangle$

5 $\langle u - v, w \rangle = \langle u, w \rangle - \langle v, w \rangle$

6 $\langle u, v - w \rangle = \langle u, v \rangle - \langle u, w \rangle$

Norm

→ Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. The norm of a vector $u \in V$ is defined by $\|u\| = \sqrt{\langle u, u \rangle}$.

→ The norm of a vector represents its length OR magnitude.

→ Let $u \in V$ be a vector in an inner product space. If $\|u\| = 1$ then u is called a **unit vector** in V.

Properties of Norm/Length

→ Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Consider $u, v \in V$ and $k \in \mathbb{R}$, then

1	$\|u\| \geq 0$	(Non-negativity)
2	$\|u\| = 0 \Leftrightarrow u = 0$	(Definiteness)
3	$\|ku\| = \|k\|\|u\|$	(Homogeneity)
4	$\|\langle u, v \rangle\| \leq \|u\|\|v\|$	(Cauchy-Schwarz Inequality)
5	$\|u + v\| \leq \|u\| + \|v\|$	(Triangle/Minkowski's inequality)
6	$\|u + v\|^2 + \|u - v\|^2 = 2\|u\|^2 + 2\|v\|^2$	(Parallelogram Law)
7	$\langle u, v \rangle = \frac{1}{4}\|u + v\|^2 - \frac{1}{4}\|u - v\|^2$	(Polarization Identity)

→ **Norm induced by the different inner product:**

1 Norm induced by the Euclidean inner product:

- For vector $u = (u_1, u_2, \ldots, u_n)$, then $\|u\| = \sqrt{\langle u, u \rangle} = \sqrt{u \cdot u} = \sqrt{(u_1, u_2, \ldots, u_n) \cdot (u_1, u_2, \ldots, u_n)} = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$.

- The norm induced by the Euclidean inner product is called the Euclidean norm.

2 Norm induced by the weighted Euclidean inner product:

- For vector $u = (u_1, u_2, \dots, u_n)$, then $\|u\| = \sqrt{\langle u, u\rangle} = \sqrt{\langle (u_1, u_2, \dots, u_n), (u_1, u_2, \dots, u_n)\rangle} = \sqrt{w_1 u_1^2 + w_2 u_2^2 + \cdots + w_n u_n^2}$.
- The norm induced by the weighted Euclidean inner product is called the weighted Euclidean norm.

Distance

→ Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space with norm $\|\cdot\|$. The distance between two vectors u and v from V is defined by $d(u, v) = \|u - v\|$.

Thus, the distance between two vectors is the norm of their difference.

Properties of distance:

→ Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. Consider $u, v, w, a \in V$ and $k \in \mathbb{R}$, then

1	$d(u, v) \geq 0$	(Non-negativity)
2	$d(u, v) = 0 \Leftrightarrow u = v$	(Definiteness)
3	$d(u, v) = d(v, u)$	(Symmetry)
4	$d(u, w) \leq d(u, v) + d(v, w)$	(Triangle Inequality)
5	$d(u + a, v + a) = d(u, v)$	(Translation Invariance)
6	$d(ku, kv) = \|k\| d(u, v)$	(Homogeneity)

Angle between two vectors **u** & **v**

→ If u & v are two non-zero vectors from an inner product space V and if θ is an angle between u & v, then

$$\cos\theta = \frac{\langle u, v\rangle}{\|u\| \cdot \|v\|} \quad \textbf{OR} \quad \theta = \cos^{-1}\left(\frac{\langle u, v\rangle}{\|u\| \cdot \|v\|}\right).$$

Examples of Method-2: Properties of Inner Product and Norm

C	1	Let $V = \mathbb{R}^2$ with inner product defined by $\langle u, v\rangle = u_1v_1 + u_2v_2$. Let $u = \begin{pmatrix}1\\1\end{pmatrix}$ and $v = \begin{pmatrix}-1\\1\end{pmatrix}$. **1.** Find the length of u and v. **2.** Find distance between u and v. **3.** Find angle between u and v. **4.** Verify triangle inequality. **5.** Verify Cauchy-Schwartz inequality. **Answer:** $(\sqrt{2}, \sqrt{2})$, **2**, $\frac{\pi}{2}$
C	2	Let the vector space $\mathbb{P}_2$ have the inner product $\langle p, q\rangle = \int_{-1}^{1} p(x)\, q(x)\, dx$. Find $\|p\|$ for $p = x^2$ and find $d(p, q)$ if $p = 1$ and $q = x$. **Answer:** $\frac{\sqrt{2}}{\sqrt{5}}$, $\frac{2\sqrt{2}}{\sqrt{3}}$
C	3	Verify Parallelogram law for $\begin{bmatrix}-1 & 2\\6 & 1\end{bmatrix}$ and $\begin{bmatrix}1 & 0\\3 & 2\end{bmatrix}$ under the Euclidean inner product on $\mathbb{M}_{22}$.
C	4	Let V be an inner product space. For $u, v \in V$ prove that $\langle u, v\rangle = \frac{1}{4}\|u + v\|^2 - \frac{1}{4}\|u - v\|^2$ (**Polarization Identity**).

Method 3 ⇝ Orthogonal and Orthonormal Sets

Orthogonal and Orthonormal Vectors

→ Two vectors u and v are said to be **orthogonal** when they are perpendicular. It means $\langle u, v\rangle = 0$, i. e. $\theta = \frac{\pi}{2}$.

→ Two vectors u and v are said to be **orthonormal** when they are unit vectors and perpendicular.

It means $\|u\| = \|v\| = 1$ and $\langle u, v\rangle = 0$, i. e. $\theta = \frac{\pi}{2}$.

→ If any two non-zero vectors are orthogonal/orthonormal then they are linearly independent.

Orthogonal and Orthonormal Set

→ A set $S = \{u_1, u_2, \dots, u_n\}$ of vectors in an inner product space V is called an orthogonal set if each pair of distinct vectors in S are orthogonal. i. e. , $\langle u_i, u_j\rangle = 0$ for $i \neq j$.

→ An orthogonal set $S = \{u_1, u_2, \dots, u_n\}$ with unit vectors is called orthonormal set.

i.e., $\langle u_i, u_j\rangle = 0$ for $i \neq j$ and $\langle u_i, u_i\rangle = 1$ for $i = 1,2, \dots, n$.

Remarks:

1 Any orthogonal set of n–non zero vectors in $\mathbb{R}^n$ is a basis for $\mathbb{R}^n$.

2 In an inner product space, a basis contains orthogonal vector is called orthogonal basis.

3 In an inner product space, a basis contains orthonormal vector is called orthonormal basis.

→ **Pythagorean theorem:** The vectors u and v are orthogonal (perpendicular) vectors in a **real** inner product space V if and only if $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

→ If u is any nonzero vector then the unit vector in the direction of u is given by $\hat{u} = \frac{u}{\|u\|}$.

→ The process of dividing non-zero vector u by its norm is called normalizing u.

Examples of Method-3: Orthogonal and Orthonormal Vectors

C	1	Determine whether the given vectors are orthogonal with respect to the Euclidean inner product. **1.** $u = (-1, \ 3, \ 2)$ and $v = (4, \ 2, \ -1)$. **2.** $u = (-4, \ 6, \ -10, \ 1)$ and $v = (2, \ 1, \ -2, \ 9)$. **Answer: (1) Orthogonal, (2) not orthogonal**
C	2	For what values of a and b is the set {u, v} orthogonal with respect to the Euclidean inner product, where $u = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$ & $v = \left(a, \frac{1}{\sqrt{2}}, -b\right)$. **Answer: $a = b$ and $a, b \in \mathbb{R}$**
C	3	Find two vectors in $\mathbb{R}^2$ with Euclidean norm 1 & whose Euclidean inner product with $(-3, \ 1)$ is 0. **Answer: $\pm \frac{1}{\sqrt{10}}(1, \ 3)$**
C	4	For two vectors a and b, $\|a + b\| = \|a - b\|$. Find the angle between a and b. **Answer: $\frac{\pi}{2}$**
C	5	Find an orthogonal basis for $\mathbb{R}^3$ containing the vectors $u = (2, -2, 1)$, $v = (2, 1, -2)$ using the Euclidean inner product and hence convert in orthonormal basis. **Answer: $w = (1, 2, 2)$ and $\left\{\frac{u}{3}, \frac{v}{3}, \frac{w}{3}\right\}$**

Method 4 ⇝ Gram-Schmidt Orthogonalization

Orthogonal Projection

→ Let u & v be two vectors in an inner product space & $v \neq \mathbf{0}$ then orthogonal projection of u along v is $\text{proj}_v u = \frac{\langle u, v\rangle}{\langle v, v\rangle} v = \frac{\langle u, v\rangle}{\|v\|^2} v$.

→ **Remarks:**

1 $\text{proj}_v u$ is orthogonal to $u - \text{proj}_v u$.

2 v is orthogonal to $u - \text{proj}_v u$.

Gram Schmidt Orthogonalization Process

→ Let V be any non-zero n-dimensional inner product space and $S_1 = \{u_1, u_2, \ldots u_n\}$ is an arbitrary basis for V. Then procedure to find orthogonal basis $S_2 = \{v_1, v_2, \ldots, v_n\}$ from S_1 is as follow:

1 Let $v_1 = u_1$.

2 Find $v_2, \ldots, v_n$ by formula

$$v_i = u_i - \frac{\langle u_i, v_1\rangle}{\|v_1\|^2} \cdot v_1 - \frac{\langle u_i, v_2\rangle}{\|v_2\|^2} \cdot v_2 - \cdots - \frac{\langle u_i, v_{i-1}\rangle}{\|v_{i-1}\|^2} \cdot v_{i-1}, \text{where } i = 2,3, \ldots, n.$$

3 The set S_2 of vectors $v_1, v_2, \ldots, v_n$ is an orthogonal set.

4 The orthogonal basis S_2 can be converted to orthonormal basis by normalizing all vectors of S_2.

Examples of Method-4: Gram-Schmidt Orthogonalization

C	1	Find the orthogonal projection of $u = (4, 0, -1)$ along $v = (3, 1, -5)$ in $\mathbb{R}^3$ with respect to the Euclidean inner product. **Answer:** $\left(\frac{51}{35}, \frac{17}{35}, -\frac{17}{7}\right)$
C	2	Let $u = \begin{pmatrix}1\\3\end{pmatrix}, v = \begin{pmatrix}1\\1\end{pmatrix}$, then find $\text{proj}_v u$, $u - \text{proj}_v u$, also verify that $\text{proj}_v u$ is orthogonal to $u - \text{proj}_v u$. **Answer:** $\begin{pmatrix}2\\2\end{pmatrix}$ **and** $\begin{pmatrix}-1\\1\end{pmatrix}$

C	3	Let $\mathbb{R}^2$ have the Euclidean inner product. Use Gram Schmidt process to transform the basis vectors $u_1 = (1, -3), u_2 = (2, 2)$ into an orthonormal basis. **Answer:** $\left(\frac{1}{\sqrt{10}}, -\frac{3}{\sqrt{10}}\right), \left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right)$
C	4	Let $\mathbb{R}^3$ have the Euclidean inner product. Transform the basis $\{(1, 1, 1), (1, -2, 1), (1, 2, 3)\}$ into an orthonormal basis using the Gram Schmidt process. **Answer:** $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right), \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$

Method 5 ⇝ QR Decomposition

QR Decomposition

→ QR decomposition is a matrix factorization technique used to express a matrix as the product of two simpler matrices.

→ QR decomposition is widely used in solving systems of linear equations, least squares problems, eigenvalue computations, and numerical analysis.

→ Let $A = [a_{ij}]_{m\times n}$ be a real $m \times n$ matrix whose columns are linearly independent. Then there exist an $m \times n$ matrix Q with orthonormal columns and an $n \times n$ upper triangular matrix R such that $A = QR$. Moreover, the diagonal entries of R are positive.

Procedure of QR Decompostion of a Matrix A

→ Let $A = [a_{ij}]_{m\times n}$ be a real $m \times n$ matrix whose columns are linearly independent.

1. Express the matrix A in terms of its column vectors: $A = [C_1 \quad C_2 \quad \dots \quad C_n]$, where $C_1, C_2, \dots, C_n$ are the column vectors of A.
2. Apply Gram-Schmidt orthogonalization process to the column vectors $C_1, C_2, \dots, C_n$ to obtain orthonormal vectors $C_1', C_2', \dots, C_n'$.
3. Form the orthonormal matrix $Q = [C_1' \quad C_2' \quad \dots \quad C_n']$. i.e., $Q^TQ = I_n$.
4. Compute $R = Q^TA$.

Examples of Method-5: Orthogonal Matrices and QR Decomposition

C	1	Find the QR decomposition of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$. **Answer:** $\begin{bmatrix} \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \end{bmatrix}$ & $\begin{bmatrix} \frac{3}{\sqrt{3}} & \frac{2}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$

C	2	Find the QR decomposition of $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 4 \end{bmatrix}$ and verify that $QR = A$. **Answer:** $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix}$ & $\begin{bmatrix} \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} \end{bmatrix}$

Method 6 ⇝ Least Squares Solutions

Least Squares Solution

→ A linear system $AX = B$ is consistent if it has unique/infinite solution and it is inconsistent if it has no solution. In many situations an inconsistent system requires a solution. Then we have to use least squares solution.

→ Given a linear system $AX = B$ of m equations and n unknowns, find a vector X, if possible, that minimizes $\|AX - B\|$ with respect to the Euclidean inner product on $\mathbb{R}^m$. Such a vector is called a least squares solution of $AX = B$.

→ For any linear system $AX = B$, the associated normal system $A^TA\widehat{X} = A^TB$ is consistent, and all solutions of the normal system are least squares solutions of $AX = B$.

→ If W is the column space of A, and $\widehat{X}$ is any least squares solution of $AX = B$, then the orthogonal projection of B on W is $\text{Proj}_W B = \widehat{B} = A\widehat{X}$, which is important to observe error vector and least square error.

Procedure for finding the least squares solution for $AX = B$

1. If the system of equations is not given directly, first construct the matrix equation $AX = B$ from the given data.
2. Compute the matrices A^TA and A^TB.
3. Solve the normal equations $A^TA\widehat{X} = A^TB$ for the unknown vector $\widehat{X}$ (least square solution) using Gauss Elimination method (or any suitable method).

→ Note:

- Compute $\widehat{B} = A\widehat{X}$, which is orthogonal projection B onto the column space of A.

Examples of Method-6: Least Squares Solutions

C	1	Find the least square solution of $AX = B$ given by $x_1 - x_2 = 4, 3x_1 + 2x_2 = 1, -2x_1 + 4x_2 = 3$ and find the orthogonal projection of B onto the column space of A. **Answer:** $\begin{bmatrix} \frac{17}{95} \\ \frac{143}{285} \end{bmatrix}, \begin{bmatrix} -\frac{92}{285} \\ \frac{439}{285} \\ \frac{94}{57} \end{bmatrix}$
C	2	Let $A = \begin{bmatrix} -2 & 3 \\ 1 & -2 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$. Find the least square solution to $AX =$ B. Find the orthogonal projection of B onto column space of A. **Answer:** $\begin{bmatrix} \frac{13}{3} \\ 3 \end{bmatrix}, \begin{bmatrix} \frac{1}{3} \\ -\frac{5}{3} \\ \frac{4}{3} \end{bmatrix}$

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